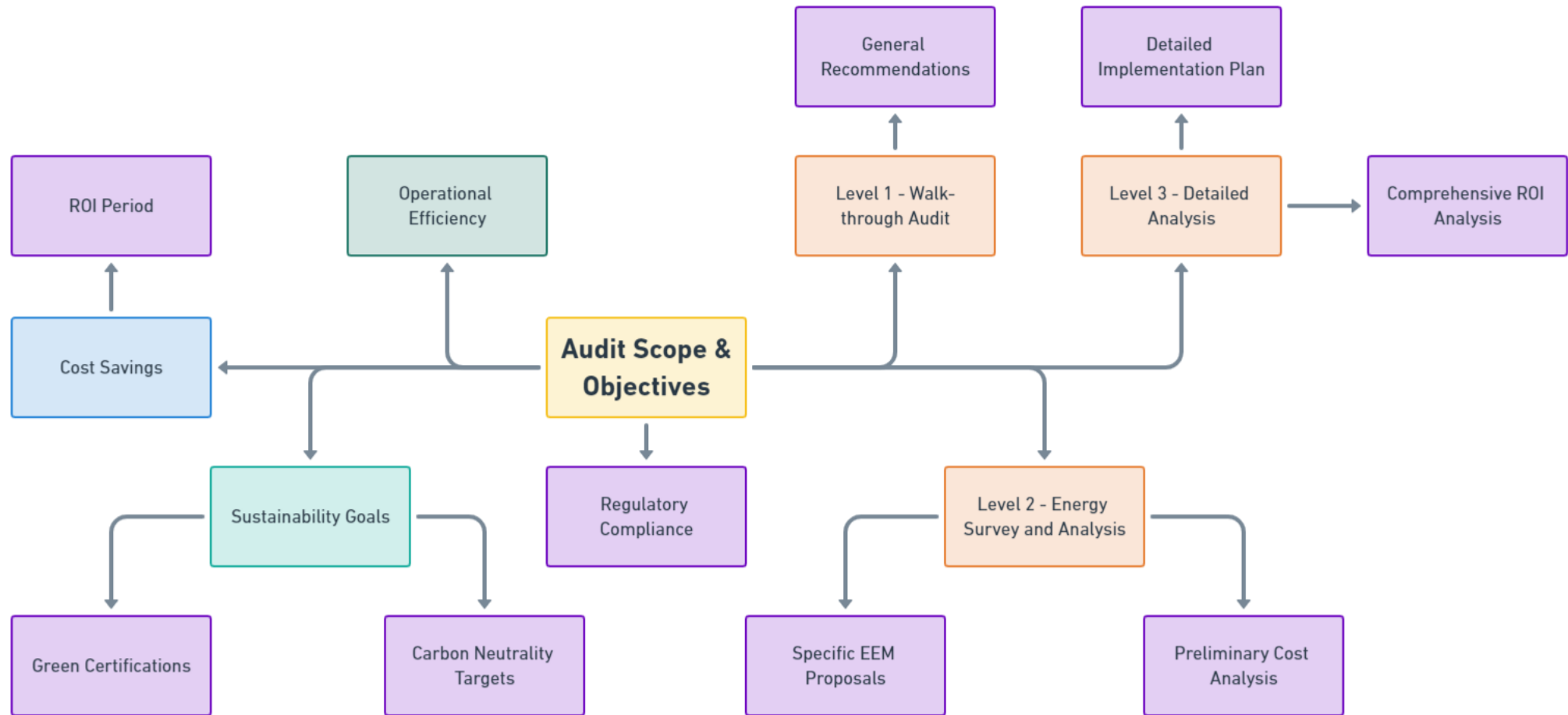
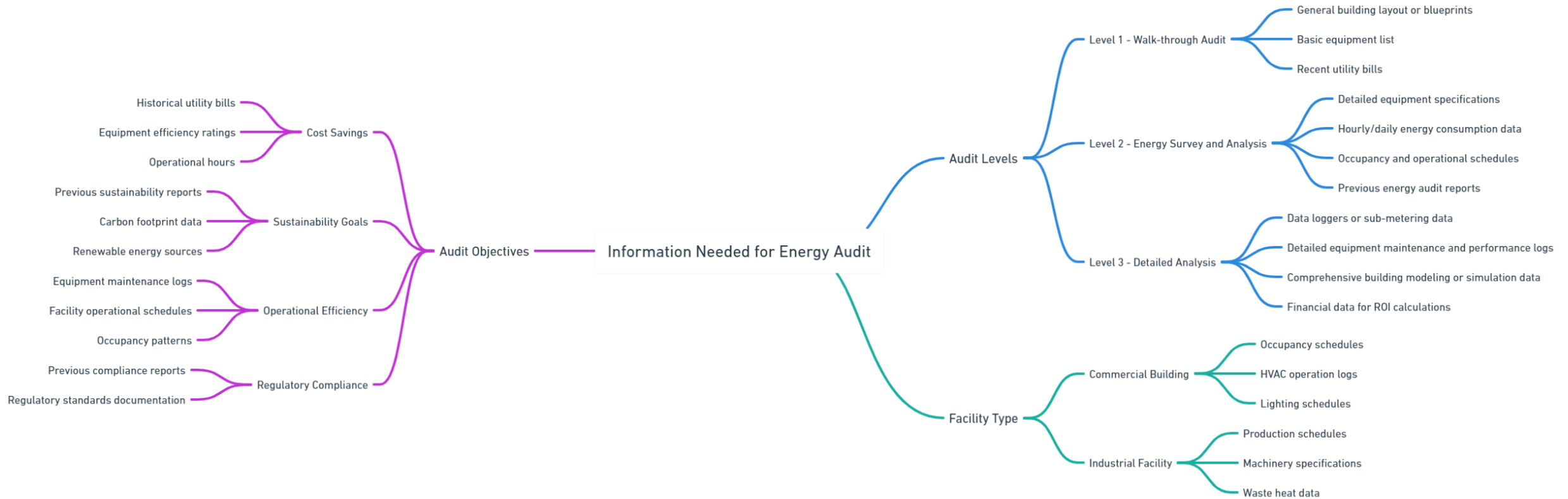


Planning the Audit



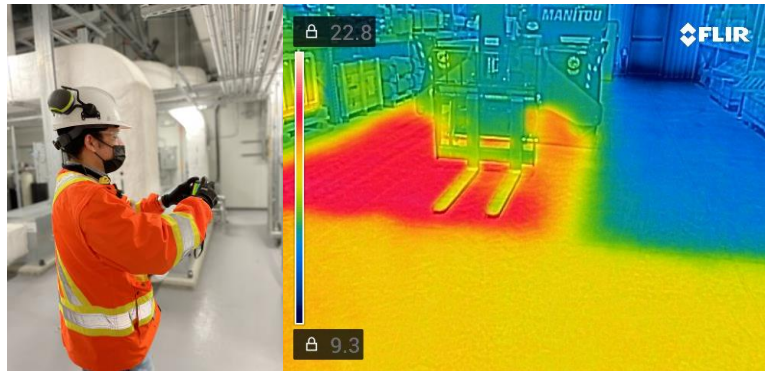
Made with Whimsical

Planning the Audit



Made with Whimsical

Planning the Audit



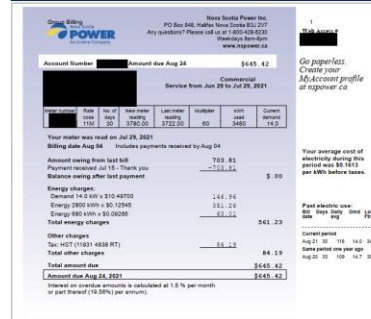
TOOLS FOR THE
ENERGY AUDIT



Conducting the Audit

- Site Visits
 - Stick to your plan... unless reality hits you in the face!

Documentation



SAFETY toolbox meeting



Preliminary Walkthrough

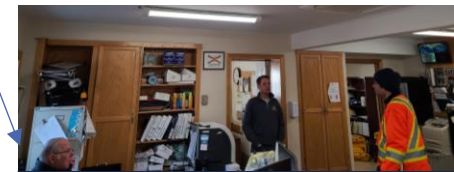
Equipment Checks



Data Collection



Interviews



Conducting the Audit

RETScreen Expert - Digby - 5 Bay Garage - NSDPW - COMBINED.rctx

File Location Facility Energy Cost Emission Finance Risk Report Custom

Benchmark Feasibility Performance All Level 1 Level 2 Hide benchmark Benchmark database... Show notes Show image Default image Select image... Adjust image... Keep aspect ratio Help eLearning Workflow... Colour coding...

Step 1 - Analysis type Step 2 - Benchmark Options Help

Subscriber: Hysovent Limited - Professional

RETScreen - Facility

Facility information

Facility type: Commercial/Institutional
Type: Other
Description: Commercial/Institutional - Maintenance Shop

Prepared for: Nova Scotia Department of Public Works
Prepared by: Hysovent Limited

Facility name: Digby - 5 Bay Garage
Address: 200 Robinson Weir Rd
City/Municipality: Digby
Province/State: NS
Country: Canada

Benchmark - Commercial/Institutional - Other

Facility size: 589 m²

Fuel consumption

Energy use intensity: GJ/m²
Reference unit: m²

Benchmark: 0.82 GJ/m²
Minimum (Typical): 0.62 GJ/m²
Maximum (Typical): 2.16 GJ/m²

Base case: 1.44 GJ/m²
Reference year: 2022

Set target: Target
Year: 2030
Target: -50%
Proposed case: 0.72 GJ/m²

Facility - Plan Annual

Energy use intensity (GJ/m²)

Plan

Base case

Target

Benchmark

Net zero

Let's go with Live Example!

- Data Analysis – RETScreen + Excel + ASHRAE Forms

All Audit Levels - Annual Summary

Summary for Existing Building

user input
calculated

Imported Energy (E_{imp})

Energy Type	Annual	Units	Conversion	Mass	Annual Cost
Electricity	47,071	kWh	3.4	172,235	\$ 7,873
	-				\$ -
	-				\$ -
	-				\$ -
	-				\$ -
Oil	17,276	Barrel (Fuel Oil #2)	39.3	672,441	\$ 15,211
	-				\$ -
	-				\$ -
E_{imp} Total					\$ 23,084

On-Site Renewable Energy Production (E_{rs})

Energy Type	Annual	Units	Conversion	Mass	Annual Cost
	-				\$ -
	-				\$ -
E_{rs} Total					\$ -

Exported Energy (E_{exp})

Energy Type	Annual	Units	Conversion	Mass	Annual Cost
	-				\$ -
	-				\$ -
E_{exp} Total					\$ -

Existing Building EUECI

Building Name	Digby 5 Bay Garage
Greenhouse Gas Emissions	898
EUI (Building) (kWh/m ² /yr)	1,418.2
EUI (Site) (kWh/m ² /yr)	1,418.2
Site ECI (Energy cost index \$/m ² /yr)	\$ 44.41

EUI: Energy Use Intensity

Benchmarking & Target EUECI

Item	Value
Benchmark Source	
Benchmark EUI (Multi/m ² /yr)	
Target EUI (Building) (Multi/m ² /yr)	
Target EUI (Site) (Multi/m ² /yr)	
Target ECI (Site) (\$/m ² /yr)	
Annual Energy Savings to Reach Target	14
Annual Cost Savings to Reach Target	(\$ available)

Notes

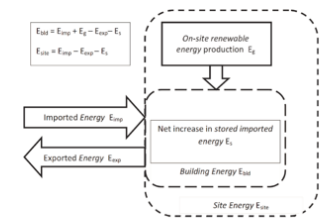
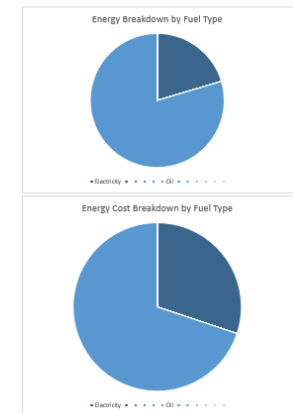


Figure 1 Relationship of site energy (E_{site}) and building energy (E_{bd})